

Department of Biostatistics and Epidemiology  
Rutgers School of Public Health  
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## Education

|             |                                                               |               |
|-------------|---------------------------------------------------------------|---------------|
| 2016 - 2020 | UNIVERSITY OF MICHIGAN<br>Ph.D. Biostatistics                 | Ann Arbor, MI |
| 2014 - 2016 | UNIVERSITY OF MICHIGAN<br>M.S. Biostatistics                  | Ann Arbor, MI |
| 2009 - 2013 | HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY<br>B.S. Biology | Wuhan, China  |

## Research Interests

Online statistical inference, epigenetic clocks, transfer learning, mobile health, dynamic mediation analysis, longitudinal data analysis, causal inference, adaptive experimental design

## Professional Appointments

|                |                                                                                                                                           |                |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2023 - Present | <b>Assistant Professor</b><br>Department of Biostatistics and Epidemiology<br>School of Public Health<br>Rutgers University               | Piscataway, NJ |
| 2020 - 2023    | <b>Assistant Professor</b><br>Department of Statistics and Actuarial Science<br>College of Liberal Arts and Science<br>University of Iowa | Iowa City, IA  |

## Teaching Experience

|                    |                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 - 2025 Fall   | <b>Instructor</b> , <i>Biostatistics Theory I</i><br>Department of Biostatistics & Epidemiology, Rutgers University                      |
| 2023 Spring        | <b>Instructor</b> , <i>Applied Linear Regression</i><br>Department of Statistics & Actuarial Science, University of Iowa                 |
| 2021 Fall          | <b>Instructor</b> , <i>Introduction to Data Science (new course)</i><br>Department of Statistics & Actuarial Science, University of Iowa |
| 2020 - 2022 Fall   | <b>Instructor</b> , <i>Mathematical Statistics I</i><br>Department of Statistics & Actuarial Science, University of Iowa                 |
| 2021 - 2023 Spring | <b>Instructor</b> , <i>Mathematical Statistics II</i><br>Department of Statistics & Actuarial Science, University of Iowa                |

## Publications

1. Han, R.\*, **Luo, L.\***, Luo, Y.\*, Lin, Y., and Huang, J. (2026). Adaptive debiased lasso in high-dimensional generalized linear models with streaming data. *Journal of the American Statistical Association (Theory & Methods)* (accepted).

2. Chen, H., Chung, L. P., Chen, Y., Balias, P., Budhkar, A. S., Punater, P., Fu, M., Schleyer, T., DeMeritt, K., Halm, E. A., Chow, L. S., **Luo, L.**, Wolfson, J. (2026). Association between therapeutic inertia and future hypoglycemia among patients with type 2 diabetes. *Diabetes Epidemiology and Management* (accepted).
3. **Luo, L.<sup>#</sup>**, Shang, L., Goodrich, J., Peterson, K., and Song, P.X.K. (2026). Bridging the gap: Enhancing the generalizability of epigenetic clocks through transfer learning. *The Annals of Applied Statistics*, **20**(1), 68–89.
4. Li, Y.<sup>†</sup>, Goodrich, J., Peterson, K., Song, P.X.K. and **Luo, L.<sup>#</sup>** (2025). Uncertainty quantification in epigenetic clocks via conformalized quantile regression. *Genetic Epidemiology*, **49**(4), e70008.
5. Ghosh, J., Tan, A. and **Luo, L.** (2025). Online Bayesian variable selection and Bayesian model averaging for streaming data. *Stat*, **14**(1), e70044.
6. Liu, W., Li, G., Zhou, L. and **Luo, L.** (2025). High-dimensional large-scale mixed-type data imputation under missing at random. *Science China Mathematics*, **68**(4), 969–1000.
7. **Luo, L.<sup>\*</sup>**, Shi, C.<sup>\*</sup>, Wang, J.<sup>\*</sup>, Wu, Z. and Li, L. (2025). Multivariate dynamic mediation analysis under a reinforcement learning framework. *The Annals of Statistics*, **53**(1), 400–425.
8. Li, Z., **Luo, L.**, Wang, J. and Feng, L. (2025). Efficient quantile covariate adjusted response adaptive experiments. *The Journal of Econometrics*, **249**, 105857.
9. **Luo, L.**, Risk, M. and Shi, X. (2024). Online causal inference with application to near real-time post-market vaccine safety surveillance. *Statistics in Medicine*, **43**(14), 2734–2746.
10. Han, R.<sup>\*</sup>, **Luo, L.<sup>\*</sup>**, Lin, Y. and Huang, J. (2024). Online inference with debiased stochastic gradient descent. *Biometrika*, **111**(1): 93–108.
11. Tran, L., Li, G., **Luo, L.** and Jiang, H. (2023). A fast solution to the lasso problem with equality constraints. *The Journal of Computational and Graphical Statistics*, **33**(3), 804–813.
12. **Luo, L.<sup>\*</sup>**, Han, R.<sup>\*</sup>, Lin, Y. and Huang, J. (2023). Online inference in high-dimensional generalized linear models with streaming data. *Electronic Journal of Statistics*, **17**(2), 3443–3471.
13. Liu, W., **Luo, L.<sup>#</sup>** and Zhou, L. (2023). Online missing value imputation for high-dimensional mixed-type data via generalized factor models. *Computational Statistics and Data Analysis*, **187**, 107822.
14. **Luo, L.**, Wang, J. and Hector, E.C. (2023). Rejoinder: ‘Statistical inference for streamed longitudinal data’. *Biometrika*, **110**(4): 871–874.
15. **Luo, L.**, Wang, J. and Hector, E.C. (2023). Statistical inference for streamed longitudinal data. *Biometrika* **110**(4), 841–858. (Selected as a discussion paper.)
16. **Luo, L.**, Zhou, L. and Song, P.X.K. (2022). Real-time regression analysis of streaming clustered data with possible abnormal data batches. *Journal of the American Statistical Association (Theory & Methods)*, **118**(543), 2029–2044.

17. **Luo, L.** and Li, L. (2022). Online two-way estimation and inference via linear mixed-effects models. *Statistics in Medicine*, **41**(25), 5113–5133.
18. Hector, E.C.\*, **Luo, L.\*** and Song, P.X.K. (2022). Parallel-and-stream accelerator for computationally fast supervised learning. *Computational Statistics and Data Analysis*, **177**, 107587.
19. **Luo, L.** and Song, P.X.K. (2021). Multivariate online regression analysis with heterogeneous streaming data. *The Canadian Journal of Statistics*. **51**(1), 111–133. (Top 10 most-cited article in 2022-2023.)
20. **Luo, L.** and Song, P.X.K. (2020). Renewable estimation and incremental inference in generalised linear models with streaming datasets. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **82**, Part1, 69–97. (An earlier version of this paper “real-time regression analysis of streaming health datasets” won the 2019 ENAR Distinguished Student Paper Award.)
21. **Luo, L.**, She, X., Cao, J., Zhang, Y., Li, Y. and Song, P.X.K. (2019). Detection and prediction of ovulation time from body temperature measured by an in-ear wearable thermometer. *IEEE Transactions on Biomedical Engineering*, **67**(2): 512–522.
22. Shen, R., **Luo, L.** and Jiang, H. (2017). Identification of gene pairs through penalized regression subject to constraints. *BMC Bioinformatics*, **18**: 466.
23. Yang, Y.\*, **Luo, L.\***, Xu, J.\* et al. (2013). Novel EDA p.Ile260Ser mutation linked to non-syndromic hypodontia. *Journal of Dental Research*, **92**: 500–506.

#### Preprints

24. Kinkade, C., Wimbish, A., Senthilkumar, A., Coker, M., **Luo, L.**, Yin, M., Kuhn, L., Corley, M., Shiau, S. (2025). Epigenetic aging in people with HIV across the life course: a systematic review.
25. Lu, Y., **Luo, L.#**, and Gu, T.# (2025). ReFIT: Federated Transfer Learning for Sequential Prediction and Uncertainty Quantification Using Streaming EHR Data.
26. Benari, O., Ravona-Springer, R., **Luo, L.**, Hu, L., Mardor, Y., Lesman-Segev, O. H., Shekhtman, S. G., Livny, A., Harel, M., Almog, G., Bendlin, B. B., Sano, M., and Beeri, M. S. (2025). The effects of hyperbaric oxygen therapy on cognition and brain in older adults with type 2 diabetes and mild cognitive impairment: a randomized double-blind sham-controlled trial.
27. Zheng, S., Liu, W., **Luo, L.**, and Zhou, L. (2025). Online partial functional linear regression.

(\*: Co-first author; #: Corresponding author; †: Student author)

## Grants

### Active

1. Addressing population and platform heterogeneity in epigenetic clocks via transfer learning and conformal prediction methods (percentile: 3%, impact score: 25)

- |         |                         |
|---------|-------------------------|
| Role:   | Principal Investigator  |
| Agency: | NIH (R01AG092615)       |
| Period: | 02/01/2026 - 01/31/2031 |
| Amount: | \$2,203,023             |
| Effort: | 37.5% FTE               |
2. Transfer learning and uncertainty quantification in epigenetic clocks
 

|         |                                                        |
|---------|--------------------------------------------------------|
| Role:   | Principal Investigator                                 |
| Agency: | NIH (R21AG083364)                                      |
| Period: | 09/30/2023 - 05/31/2025 (NCE: 06/01/2025 - 05/31/2026) |
| Amount: | \$327,576                                              |
| Effort: | 20% FTE                                                |
  3. Flexible statistical solutions for the Network of Biostatistics for RECOVER (FlexStatS – NBR)
 

|         |                         |
|---------|-------------------------|
| Role:   | Co-Investigator         |
| PI:     | Andrea Foulkes          |
| Agency: | NIH (OT2HL161841)       |
| Period: | 09/03/2024 - 08/31/2027 |
| Amount: | \$1,050,000             |
| Effort: | 15% FTE                 |
  4. Decoding the role of the glycoproteome in Alzheimer’s disease and related disorders.
 

|         |                        |
|---------|------------------------|
| Role:   | Co-Investigator        |
| PI:     | Michal Beeri           |
| Agency: | NIH (R01AG091534)      |
| Period: | 10/01/2025 - 2/28/2026 |
| Amount: | \$795,436              |
| Effort: | 15% FTE                |
  5. Cognitive and brain imaging correlates of apathy-components in asymptomatic middle aged individuals at high ADRD-risk
 

|         |                      |
|---------|----------------------|
| Role:   | Co-Investigator      |
| PI:     | Michal Beeri         |
| Agency: | NIH (R21AG080827)    |
| Period: | 3/1/2024 - 2/28/2026 |
| Amount: | \$364,709            |
| Effort: | 10% FTE              |
  6. Maternal inflammation in relation to offspring epigenetic aging and neurodevelopment
 

|         |                      |
|---------|----------------------|
| Role:   | Co-Investigator      |
| PI:     | Stephanie Shiau      |
| Agency: | NIH (R01HD111550)    |
| Period: | 6/1/2023 - 5/31/2027 |
| Amount: | \$2,628,726          |
| Effort: | 5% FTE               |
  7. Nurse-led app intervention to improve glycemic control in patients with insulin-treated type 2 diabetes on Medicaid

Role: Co-Investigator  
 PI: Helen Chen  
 Agency: NIH (R00NR020377)  
 Period: 07/01/2025 - 6/30/2028  
 Amount: \$747,000  
 Effort: 1% FTE

### Pending

1. Interaction of metabolic dysfunction and apolipoprotein E4 on longitudinal neural and cognitive outcomes

Role: Co-Investigator  
 PI: Auriel Willette  
 Agency: NIH  
 Period: 07/01/2026 - 06/30/2031  
 Amount: \$434,095  
 Effort: 10% FTE

2. The impact of ultra-processed foods on children's epigenome and neurodevelopment

Role: Co-Investigator  
 PI: Carolyn W. Kinkade  
 Agency: NIH (K99/R00)  
 Period: 07/01/2026 - 06/30/2028  
 Amount: \$197,258  
 Effort: 0% FTE

### Completed

1. Systemic and dietary advanced glycation end products in type 2 diabetes-related cognitive decline and incident dementia: effects on Alzheimer's pathology and cerebrovascular disease

Role: Co-Investigator  
 PI: Michal Beeri  
 Agency: NIH (R01AG061093)  
 Period: 12/1/2023 - 09/30/2025  
 Amount: \$1,420,317  
 Effort: 15% FTE

## **Graduate Student Mentored**

### PhD Students Advisor

2025 - Present *Kexin Guo (primary advisor)*, Department of Biostatistics & Epidemiology, Rutgers University  
 2025 Fall *Kevin Eng (project-specific mentorship)*, Department of Statistics, Rutgers University  
 2024 *Yanping Li (project-specific mentorship)*, School of Statistics and Data Science, Nankai University  
 2023 *Yoon Cho (PhD dissertation committee member)*, Department of Biostatistics, University of Iowa

### MS Students Advisor

|                |                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - Present | <i>Qilei Sheng (research assistant, co-advised with Stephanie Shiau), Department of Biostatistics &amp; Epidemiology, Rutgers University</i>                 |
| 2024 - 2025    | <i>Alexandra Lopes Ferreira, Zixuan Le (academic advisor), Department of Biostatistics &amp; Epidemiology, Rutgers University</i>                            |
| 2024 - 2025    | <i>Maria Pineda, Deen Sonbol, Tanisha Patel (MS applied practice experience advisor), Department of Biostatistics &amp; Epidemiology, Rutgers University</i> |
| 2022 - 2023    | <i>Yilin Wang (MS creative component advisor), Department of Statistics &amp; Actuarial Science, University of Iowa</i>                                      |

## Honors and Awards

|      |                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------|
| 2022 | Old Gold Summer Fellowship                                                                                   |
| 2020 | Student Paper Competition Award for Statistical Learning and Data Science                                    |
| 2020 | Winning Proposal in the 2020 Joint Shark Tank Retreat                                                        |
| 2019 | Excellence in Research Honorable Mention                                                                     |
| 2019 | Rackham Predoctoral Fellowship for Academic Year 2019-2020                                                   |
| 2019 | The International Biometric Society Eastern North American Region's (ENAR) Distinguished Student Paper Award |
| 2018 | The Michigan Institute for Data Science (MIDAS) Annual Symposium Poster Award of Most Innovative Use of Data |
| 2015 | Outstanding Academic Performance First-Year Master's Program                                                 |
| 2015 | Certificate in Public Health Genetics (CPHG) Award                                                           |
| 2013 | HUST Excellent Graduate                                                                                      |
| 2011 | China National Scholarship                                                                                   |

## Presentations

### Department - Colloquium

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <i>Uncertainty Quantification in Epigenetic Clocks via Conformalized Quantile Regression, Data Management and Modeling Core meeting of ELEMENT study, School of Public Health, University of Michigan (virtual).</i> |
| 2024 | <i>Online Statistical Inference in Streaming Data: Renewability, Dependence, and Dynamics, Department of Mathematics and Statistics, Auburn University (virtual).</i>                                                |
| 2024 | <i>Adaptive Experiments for Learning Quantile Treatment Effects, Department of Biostatistics, University of Nebraska Medical Center (virtual).</i>                                                                   |
| 2024 | <i>Adaptive Experiments for Learning Quantile Treatment Effects, Collaborative Lecture Series, Rutgers School of Public Health/Cochin University of Science and Technology (virtual).</i>                            |
| 2023 | <i>Online Statistical Inference with Streaming Data: Renewability, Dependence, and Dynamics, Department of Statistics, North Carolina State University, Raleigh, NC, United States.</i>                              |
| 2022 | <i>Online Statistical Inference in Streaming Data: Renewability, Dependence, and Dynamics, Department of Biostatistics and Epidemiology, Rutgers University (virtual).</i>                                           |
| 2022 | <i>Online Statistical Inference in Streaming Data: Renewability, Dependence, and Dynamics, Department of Statistics, North Carolina State University (virtual).</i>                                                  |

- 2022 *Online Statistical Inference with Dynamic and Dependent Streaming Data*, Data Mining Iowa Group, Business Analytics Department, University of Iowa, Iowa City, IA, United States.
- 2022 *Real-Time Regression Analysis with Dynamic Streaming Datasets*, StatScale seminars, Lancaster University, Bailrigg, Lancaster, United Kingdom (virtual).
- 2021 *Real-Time Regression Analysis with Streaming Health Datasets*, Department of Biostatistics, College of Public Health, University of Iowa, Iowa City, IA, United States (virtual).
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Statistics and Actuarial Science, University of Iowa, Iowa City, IA, United States.
- 2020 *Real-Time Regression Analysis with Streaming Health Datasets*, Public-Health Sciences – Biostatistics Program, Fred Hutchinson Cancer Research Center, Seattle, WA, United States.
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Statistics, UW Madison College of Letters & Science, Madison, WI, United States.
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Statistics and Data Sciences, College of Natural Sciences, The University of Texas at Austin, Austin, TX, United States.
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Statistics and Operations Research, The University of North Carolina at Chapel Hill, Chapel Hill, NC, United States.
- 2020 *Real-Time Regression Analysis with Streaming Health Datasets*, Department of Biostatistics, Fielding School of Public Health, University of California, Los Angeles, CA, United States.
- 2020 *Real-Time Regression Analysis with Streaming Health Datasets*, Department of Biostatistics, University of Minnesota, Minneapolis, Minnesota, United States.
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Mathematics and Statistics, McGill University, Montreal, Quebec, Canada.
- 2020 *Renewable Estimation and Incremental Inference with Streaming Health Datasets*, Department of Statistics and Actuarial Science, University of Waterloo, Waterloo, ON, Canada.

### Conference Presentation

- 2025 *Online Statistical Inference in Streaming Data: Renewability, Dependence, and Dynamics*. 2025 ISU-NISS Conference on Statistics in AI, Ames, IA, United States.
- 2025 *Online Statistical Inference with Dynamic and Dependent Streaming Data*. 2025 ENAR Spring Meeting, New Orleans, LA, United States.
- 2024 *Adaptive Experiments for Learning Quantile Treatment Effects*. The Fifth International Workshop on Statistical Analysis of Multi-Outcome Data (SAM 2024), Salzburg, Austria.
- 2023 *Sequential Monitoring and Near Real-Time Post-Market Vaccine Safety Surveillance*. 2023 ICSA Applied Statistics Symposium, Ann Arbor, MI, United States.

- 2022 *Multivariate Online Regression Analysis with Heterogeneous Streaming Data*. Joint Statistical Meetings, Washington D.C., United States.
- 2022 *Multivariate Online Regression Analysis with Heterogeneous Streaming Data*. The Fifth ICSA-Canada Chapter Symposium, Banff, Canada.
- 2021 *Multivariate Online Regression Analysis with Heterogeneous Streaming Data*. International Chinese Statistical Association (virtual).
- 2021 *Sequential Monitoring and Near Real-Time Post-Market Vaccine Safety Surveillance*. International Chinese Statistical Association (virtual).
- 2020 *Real-Time Regression Analysis of Streaming Clustered Datasets*. Joint Statistical Meetings (virtual).

## Software

R packages: glmADMM, RenewGLM, RenewQIF, OnlineCausal, OnlineDSGD (available on <https://github.com/luolsph>)

## Professional Activities and Service

### Editorial Positions

2024 - Present Associate Editor, Biometrics

### Journal Referee

Journal of the Royal Statistical Society: Series B, Nature Communications, Journal of the American Statistical Association, Biometrika, Journal of Econometrics, Journal of Machine Learning Research, Journal of Royal Statistical Society: Series A, Annals of Applied Statistics, Statistical Sinica, Biometrics, Statistics in Medicine, Electronic Journal of Statistics, Journal of Computational and Graphical Statistics, Technometrics, Journal of Nonparametric Statistics, Statistics and Its Interface, Scandinavian Journal of Statistics, Computational Statistics and Data Analysis, Statistical Analysis and Data Mining, IEEE Transactions on Neural Networks and Learning Systems, International Journal of Biostatistics

### Grant Reviews

2023 Department of Defense, Biostatistical Reviewer



### Department & University

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|----------------|---------------------------------------------------------------------------------------------------------------|
| 2025 October   | Interprofessional Education, Faculty Facilitator, Rutgers University                                          |
| 2025 - Present | Curriculum Committee, Member, Rutgers School of Public Health                                                 |
| 2025           | Biostatistics Tenure-Track Faculty Search Committee, Member, Rutgers School of Public Health                  |
| 2025 August    | Data Safety & Monitoring Committee, Biostatistician, Bright IDEAS research study, Rutgers Cancer Institute    |
| 2024 - Present | Seminar Committee, Co-chair, Department of Biostatistics and Epidemiology, Rutgers University                 |
| 2023 - Present | Statistical Health for AI Committee, Member, Department of Biostatistics and Epidemiology, Rutgers University |
| 2024           | Rutgers Cancer Institute of New Jersey (CINJ) Faculty Search Committee, Member                                |
| 2023           | Colloquium Committee, Member, University of Iowa                                                              |
| 2022 - 2023    | Executive Committee, Member, University of Iowa                                                               |
| 2021 - 2022    | Graduate Minor Exam Committee, Member, University of Iowa                                                     |
| 2020 - 2023    | Computer Committee, Member, University of Iowa                                                                |
| 2020 - 2022    | Social Committee, Chair, University of Iowa                                                                   |

### Member

|                |                                               |
|----------------|-----------------------------------------------|
| 2021 - Present | International Chinese Statistical Association |
| 2018 - Present | American Statistical Association              |
| 2017 - Present | International Biometric Society, ENAR         |

### Conference service

|      |                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | SAM 2026, Session Organizer, <i>Transfer learning in biomedical studies</i> (scheduled)                                                   |
| 2024 | JSM 2024, Session Chair, <i>New methods for correlated data</i>                                                                           |
| 2024 | SAM 2024, Session Chair, <i>Joint modeling of complex survival data</i>                                                                   |
| 2024 | SAM 2024, Session Organizer, <i>Analysis of data with multiple treatments and mixed outcomes</i>                                          |
| 2024 | SAM 2024, Committee Member on Junior Researcher Award                                                                                     |
| 2023 | ICSA Applied Statistics Symposium, Session Organizer and Chair, <i>Causal inference in observational studies and adaptive experiments</i> |
| 2023 | ICSA Applied Statistics Symposium, Student Paper Competition Committee                                                                    |
| 2023 | Joint Statistical Meetings, Statistical Learning and Data Science (SLDS) Student Paper Competition Committee                              |
| 2022 | Joint Statistical Meetings, Session Organizer and Chair, <i>Recent advances in streaming data analytics</i>                               |